

F O R T I M A N A G E R N S E 6

Lab 04 — Device Registration and FGFM

01. Registration Methods and FGFM

Labs 1 to 3 built a FortiManager and configured how administrators work on it. This lab brings a firewall under its management, which turns on the FGFM tunnel every later lab depends on.

It is documented across three attempts rather than one. The first two failed, and the reason they failed turns out to be more instructive than a clean registration would have been — so the certificate behavior that blocked them is the substance of the lab, and the successful registration at the end is the proof that the mechanics work once the blocking condition is removed.

► 1.1 The FGFM protocol

FortiGate-FortiManager protocol is the management channel between the two appliances. It runs as the `fgfmd` daemon on the FortiGate and `fgfmsd` on FortiManager, and establishes a TLS tunnel over TCP port 541.

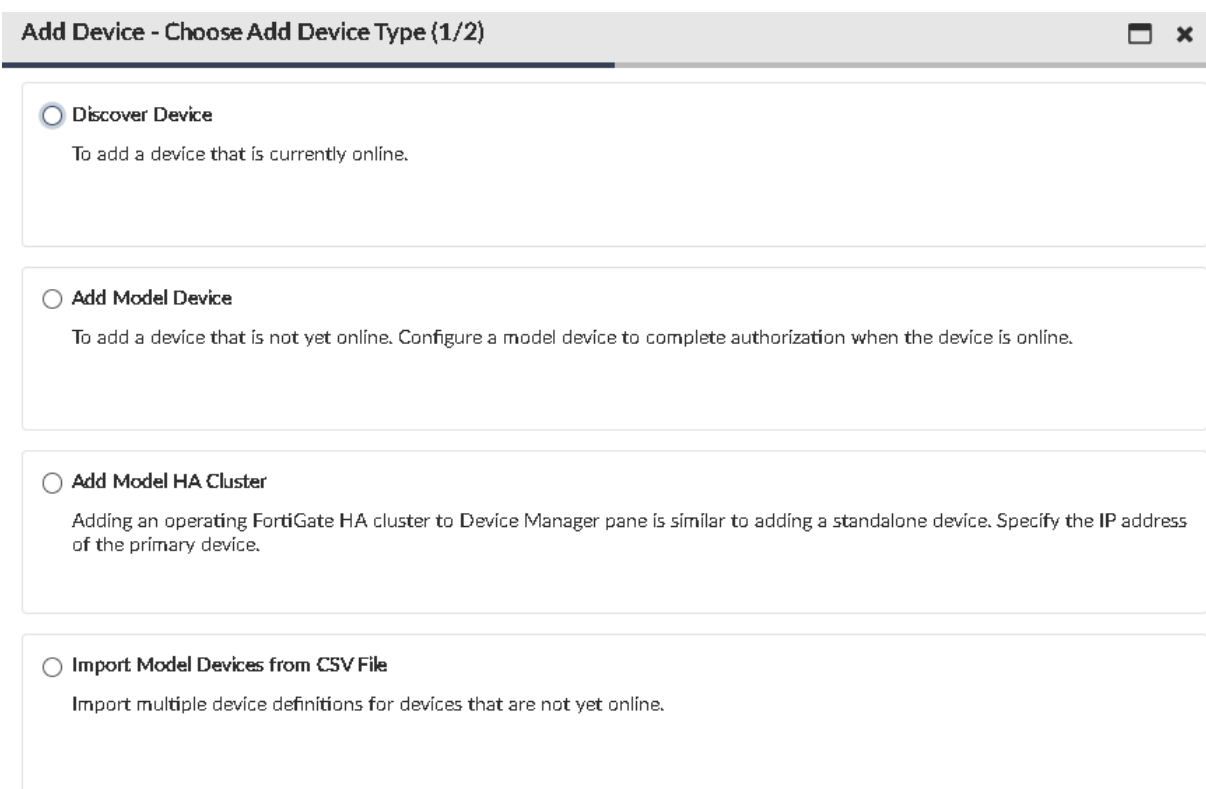
Either end can initiate it, and which end does decides what has to be configured and what has to be reachable.

- **FortiManager-initiated** — FortiManager connects outward to the FortiGate, so it needs reachability to the device and administrative credentials for it.
- **Device-initiated** — the FortiGate is pointed at FortiManager and requests the relationship. It needs only outbound reachability to port 541, which is why this is the normal method for remote sites behind NAT.

Because the session is carried over a single outbound TCP connection, FGFM survives port-based NAT in the path. That is the practical reason the device-initiated path is preferred in distributed deployments.

► 1.2 The methods FortiManager offers

Device Manager > Device & Groups > Add Device presents four options.



Add Device - Choose Add Device Type (1/2)

- ☐ **Discover Device**
To add a device that is currently online.
- ☐ **Add Model Device**
To add a device that is not yet online. Configure a model device to complete authorization when the device is online.
- ☐ **Add Model HA Cluster**
Adding an operating FortiGate HA cluster to Device Manager pane is similar to adding a standalone device. Specify the IP address of the primary device.
- ☐ **Import Model Devices from CSV File**
Import multiple device definitions for devices that are not yet online.

Figure 1 — Add Device wizard — the four device-addition types.

Method	When it applies	How it works
Discover Device	Device is already online	FortiManager probes the supplied IP with administrative credentials and reads the model, serial and firmware from the device itself. On current FortiOS this uses fabric authorization; the Use Legacy Device Login toggle covers 6.4 and older.
Add Model Device	Device does not exist yet	Zero-touch provisioning. A placeholder is created against a serial number with configuration staged in advance, and the real device adopts it on first contact.
Add Model HA Cluster	HA pair not yet online	As above for a cluster; the IP address supplied is the primary unit.
Import Model Devices from CSV	Bulk staging	Multiple model-device definitions loaded from a file.

► 1.3 Lab environment

Three device pairs were used over the course of the lab. The first is the pair carried forward from Labs 1 to 3; the second and third were built in response to the failures described in sections 03 and 04.

Pair	FortiManager	FortiGate	Outcome
A	7.6.2 build3415 — 172.16.31.183	7.0.15 (FGVMEVS4JZ695LF3) — 172.16.31.184, unlicensed	Rejected on certificate CN
B	7.6.7 build3737 — 10.58.228.84, licensed	7.6.7 build3704 (FGVMEVZ1T3HWS4B8) — 10.58.228.85, licensed	Same rejection
C	7.4.3 build2487 — FMG- VM64	7.2.1 build1254 (FGVMEV6EP-2D3VB4) — 10.58.228.177	Registered and synchronized

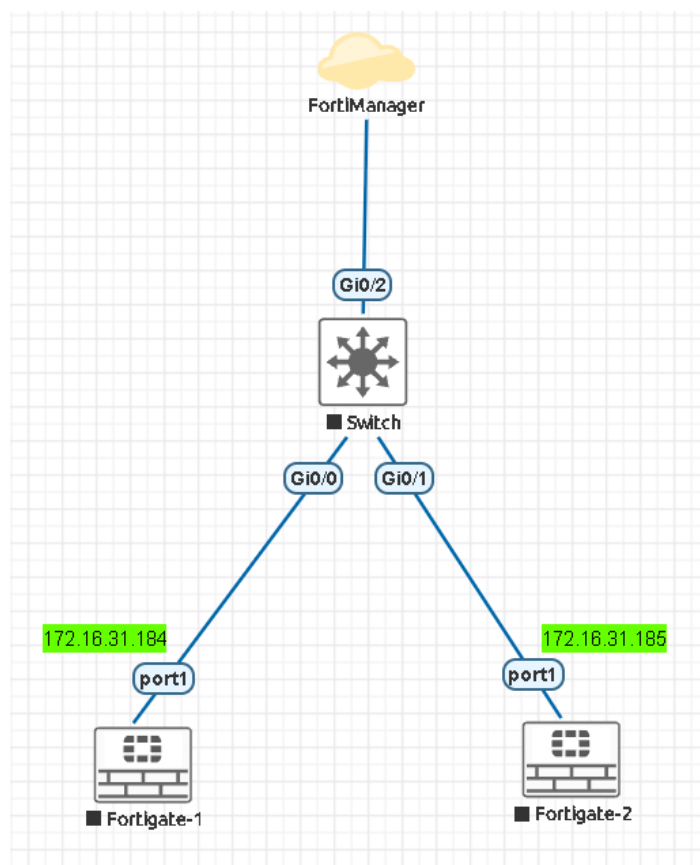


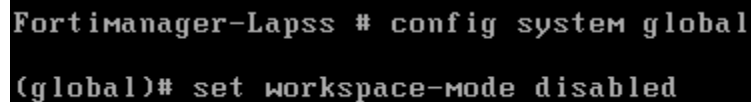
Figure 2 — Lab topology — FortiManager and two FortiGates on a shared management segment.

▸ 1.4 Prerequisites

Two FortiManager settings have to be confirmed before any registration attempt, and both were deliberately changed in earlier labs.

Workspace mode locks ADOMs for exclusive editing. Registering a device into a locked ADOM produces the read-only behaviour documented in Lab 03, where the operation appears to succeed but no change is committed.

```
config system global
    set workspace-mode disabled
end
```

A terminal window with a black background and green text. The prompt is 'Fortimanager-Lapss #'. The first command is 'config system global' and the second is '(global)# set workspace-mode disabled'.

```
Fortimanager-Lapss # config system global
(global)# set workspace-mode disabled
```

Figure 3 — Workspace mode disabled on FortiManager before registration.

Offline mode disables FGFM entirely. It was enabled in Lab 01 while the appliance was configured without upstream connectivity, and has to be turned back off here — FGFM will not establish while it is on.

Three prerequisites, one symptom

Workspace mode disabled, offline mode disabled, and fgfm present in the allowaccess list of the FortiGate interface facing FortiManager.

Any one of the three missing produces a failure that looks like a certificate or credential problem but is not.

02. Attempt 1 — Model Device on FortiManager 7.6.2

The model device path was built first, because it exercises the serial-number linkage that the rest of the lab depends on.

▸ 2.1 Capturing the serial number

A model device is matched to a real one by serial number alone, so the serial has to be read before anything is created in FortiManager. It is printed during boot.

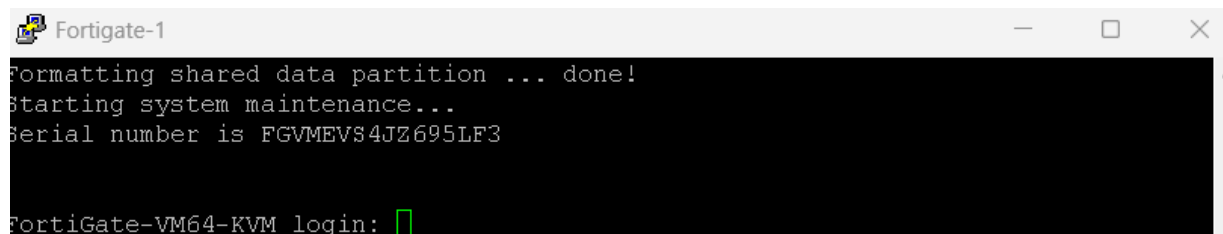


Figure 4 — FortiGate boot output showing serial number FGVMEVS4JZ695LF3.

▸ 2.2 Creating the model device



Figure 5 — Add Model Device selected in the Add Device wizard.

The device is linked by serial number rather than pre-shared key. Automatically Link to Real Device is enabled so the placeholder is adopted when the physical device makes contact, and the firmware version is left at the ADOM default.

Add Device - Provide Model Device Info (1/2)

Name

Link Device By

Serial Number
Pre-shared Key

Serial Number
FGVMEVS4JZ695LF3

Use Device Blueprint

Device Model
FortiGate-VM64

Port Provisioning
1

Has Log Disk

Automatically Link to Real Device

Enforce Firmware Version
7.6 (By Default)

Enforce Device Configuration

Managed by SD-WAN Manager

Add to Device Group

Add to Folder

Fabric Authorization Template

Pre-Run CLI Template

Assign Policy Package

Provisioning Templates
+

Metadata Variables
Edit Variable Mapping

Copy Device Dashboard
Click to select

Figure 6 — Model device definition — serial number linkage and provisioning options.

The record appears in Device Manager immediately, but in a disabled state: connectivity shows Model Device, config status Unknown, and the policy package ring is amber at Never Installed. Nothing has contacted FortiManager yet.



Figure 7 — Model device present in Device Manager with connectivity shown as Model Device.

2.3 Pointing the FortiGate at FortiManager

```

config system central-management
    set type fortimanager
    set fmg "172.16.31.183"
end

```

```

FortiGate-VM64-KVM # config system central-management
FortiGate-VM64-KVM (central-management) # set type fortimanager
FortiGate-VM64-KVM (central-management) # set fmg "172.16.31.183"
FortiGate-VM64-KVM (central-management) # end

```

Figure 8 — Central management configured on the FortiGate toward 172.16.31.183.

► 2.4 Interface access and reachability

FGFM has to be permitted on the interface facing FortiManager.

```

config system interface
  edit port1
    set mode static
    set ip 172.16.31.184 255.255.192.0
    set allowaccess fgfm http https ping
  next
end

```

```

FortiGate-VM64-KVM (port1) # set mode static
FortiGate-VM64-KVM (port1) # set ip 172.16.31.184 255.255.192.0
FortiGate-VM64-KVM (port1) # set allowaccess fgfm http https
FortiGate-VM64-KVM (port1) # set allowaccess fgfm http https ping
FortiGate-VM64-KVM (port1) # next
FortiGate-VM64-KVM (interface) # end

```

Figure 9 — Interface configuration — allowaccess issued twice, the second adding ping.

Observation	allowaccess was set twice in the capture above: the first command omits ping, the second adds it.
Why	allowaccess is a replace operation rather than an append. A shorter list silently removes services that were previously permitted.
Risk	On a management interface this can remove the administrator's own access path at the same moment it adds fgfm.

Reachability is confirmed from the FortiManager side before interpreting any registration failure.

```
execute ping 172.16....  
  
Connected  
Fortimanager-Lapss # execute ping 172.16.31.184  
PING 172.16.31.184 (172.16.31.184): 56 data bytes  
64 bytes from 172.16.31.184: seq=0 ttl=255 time=1.548 ms  
64 bytes from 172.16.31.184: seq=1 ttl=255 time=1.419 ms  
64 bytes from 172.16.31.184: seq=2 ttl=255 time=3.203 ms  
64 bytes from 172.16.31.184: seq=3 ttl=255 time=1.499 ms
```

Figure 10 — FortiManager reaching the FortiGate management address.

The FortiGate dashboard then shows the FortiManager entry with its IP populated but the status indicator red — the address is configured and the tunnel is being attempted, but the relationship is not established.

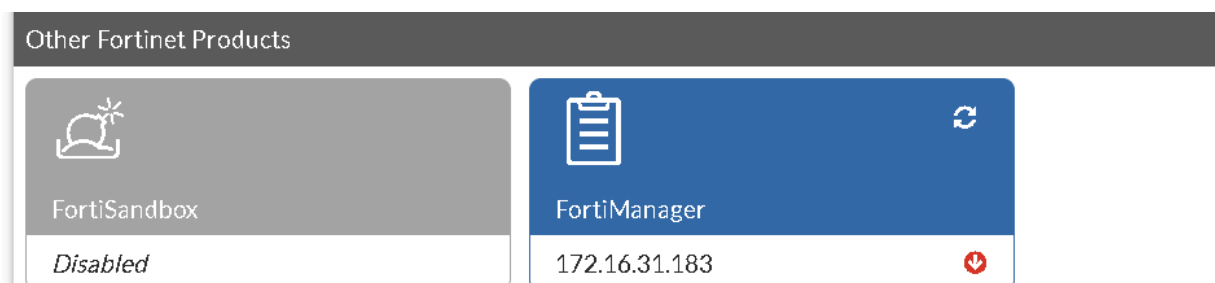


Figure 11 — FortiGate dashboard showing FortiManager configured but not connected.

Result

The FGFM tunnel reaches FortiManager and TLS negotiates. Authorization is then refused, with FortiManager logging that the serial number in the registration message does not match the subject CN of the certificate the FortiGate presented.

03. The Certificate Investigation

▸ 3.1 What FortiManager is actually checking

FGFM is mutual TLS. Both ends present a certificate, and FortiManager applies two separate tests to the one the FortiGate sends.

- **Identity** — the subject CN of the certificate must contain the device serial number, and that serial must match the one carried in the registration message.
- **Trust** — the certificate must be issued by a CA that FortiManager has been told to accept for FGFM.

In plain terms: FortiManager wants the certificate to say who the device is, and it wants that claim signed by someone it trusts. Both tests have to pass. The rejection message names only the first, which is why the problem initially looks smaller than it is.

The FortiGate holds two candidate certificates, and each passes one test and fails the other.

Certificate	Subject CN	Issuer	Identity	Trust
Fortinet_Factory	FortiGate	fortinet-subca2001	Fail	Pass
Fortinet_SSL	FGVMEVS4JZ695LF3	FGVMEVS4JZ695LF3 (self-signed)	Pass	Fail

This is the core finding of the lab. On a licensed, entitled FortiGate the factory certificate carries the serial in its CN and is signed by a Fortinet sub-CA, so both tests pass and registration is unremarkable. On an evaluation VM no reconfiguration of the two existing certificates can satisfy both tests at once.

▸ 3.2 Swapping to Fortinet_SSL

The first attempt was simply to present the certificate that does carry the serial.

```
config system central-management
    set local-cert Fortinet_SSL
end

get system central-management | grep local-cert
```

```
FortiGate-VM64-KVM # config system central-management

FortiGate-VM64-KVM (central-management) # set local-cert Fortinet_SSL

FortiGate-VM64-KVM (central-management) # end

FortiGate-VM64-KVM # get system central-management | grep local-cert
local-cert          : Fortinet_SSL
```

Figure 12 — Fortinet_SSL selected as the FGFM local certificate on the FortiGate.

The failure moved rather than cleared. FortiManager no longer complains about the CN; it refuses the certificate chain instead, because a self-signed certificate is not issued by any CA it trusts for FGFM. Identity satisfied, trust lost.

► 3.3 Building a certificate that satisfies both tests

If no supplied certificate passes both tests, the remaining option is to create one: a private CA, registered with FortiManager for FGFM, issuing a device certificate whose CN is the FortiGate serial.

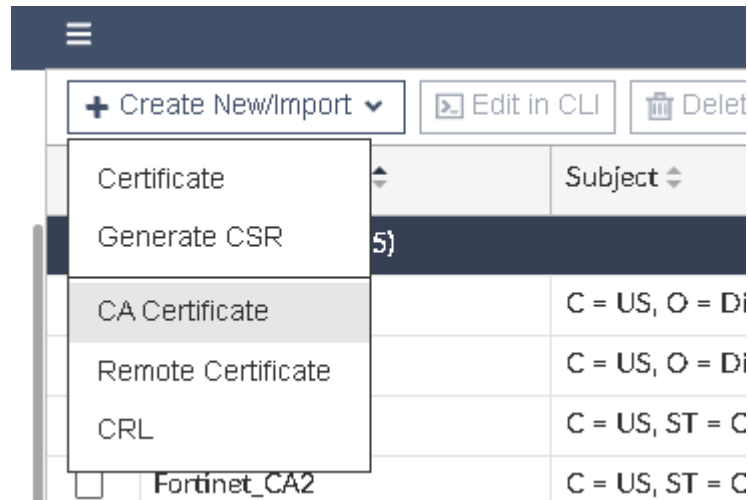


Figure 13 — Create New / Import > CA Certificate on FortiManager.

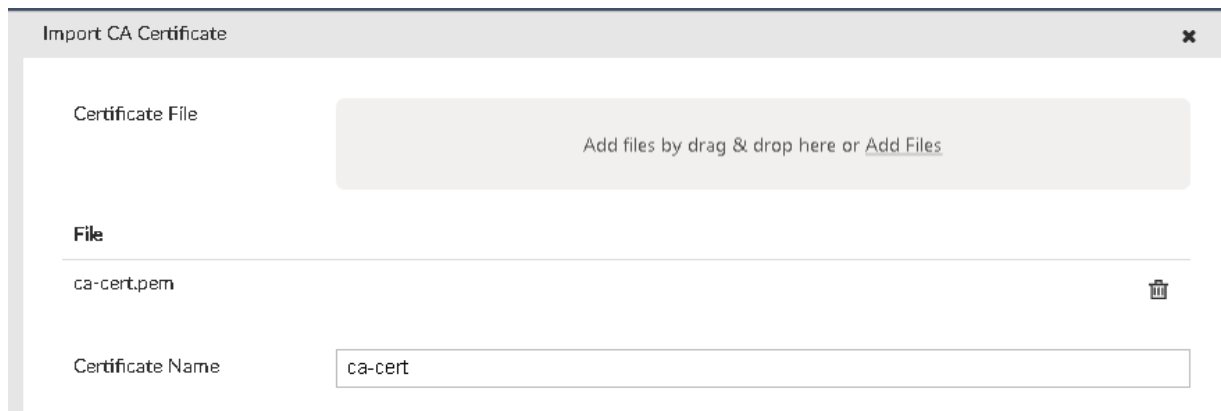


Figure 14 — Importing the lab CA as ca-cert.

The imported CA appears in the local CA list with its subject and issuer identical, confirming it is a root.



Figure 15 — Lab CA imported — CN = lab-fgfm-ca, status OK.

▸ 3.4 The key-size ceiling

The device certificate then has to be imported on the FortiGate, and this is where the evaluation VM intervenes. Keys at 2048 bits were refused outright, with the FortiGate reporting an incorrect key size for the certificate type. Progressively smaller keys were tried; only 512 bits was accepted.

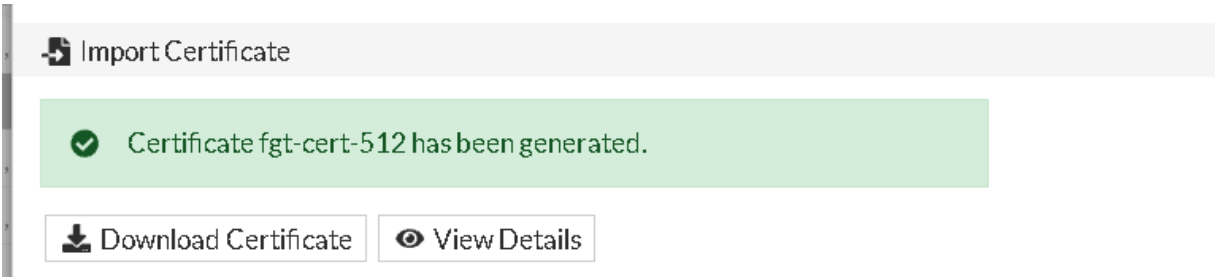


Figure 16 — Device certificate generated at 512 bits after larger keys were refused.

```
FortiGate-VM64-KVM # get system central-management | grep local-cert
local-cert          : fgt-cert-512
```

Figure 17 — FortiGate presenting the 512-bit certificate for FGFM.

Observation	Imported keys above 512 bits were refused, while the device's own factory certificates at 1024, 2048 and 4096 bits continued to work.
Why	The evaluation licence caps the size of imported certificate material. The appliance's own certificates are unaffected, because the cap applies to imported keys only.
Related	This is the same class of restriction as the reduced cipher set and low encryption algorithm seen on unlicensed VMs in NSE 4 Lab 09.

▸ 3.5 Registering the CA for FGFM

Importing a CA into the certificate store is not the same as trusting it for FGFM. FortiManager keeps a separate setting for that, and until it is set the CA is present but unused.

```
config system global
    set fgfm-ca-cert ca-cert
end
```

```

Fortimanager-Lapss # config system global

(global)# set fgfm-ca-cert
Note: "" means that default CA will be used.
ca-cert
Fortinet_CA
Fortinet_CA2
Fortinet_SUBCA
Digicert_CA
Digicert_TSA_CA

(global)# set fgfm-ca-cert ca-cert

```

Figure 18 — Selecting the lab CA as the FGFM CA on FortiManager.

The effect is visible in the FGFM debug, where the list of CA issuers FortiManager broadcasts to the peer grows to include the lab CA alongside the three Fortinet defaults.

```

__info_callback,993: role=svr,state=22, TLSv1.3 SSLv3/TLS write server hello
__info_callback,993: role=svr,state=35, TLSv1.3 SSLv3/TLS write change cipher spec
__info_callback,993: role=svr,state=37, TLSv1.3 TLSv1.3 write encrypted extensions
    CA issuer to broadcast: lab-fgfm-ca
    CA issuer to broadcast: support
    CA issuer to broadcast: fortinet-ca2
    CA issuer to broadcast: fortinet-ca2
Svr broadcast 4 CA subject names to peer
__info_callback,993: role=svr,state=25, TLSv1.3 SSLv3/TLS write certificate request
Remote CA subject is /C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=fortinet-subca2003.
issuer matching...try next if not match... local_issuer(fortinet-subca2001), remote_CA_subject(fortinet-subca2003)
Remote CA subject is /C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=support/emailAddress=support@fortinet.com.
issuer matching...try next if not match... local_issuer(fortinet-subca2001), remote_CA_subject(support)
Remote CA subject is /C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=fortinet-ca2/emailAddress=support@fortinet.com.
issuer matching...try next if not match... local_issuer(fortinet-subca2001), remote_CA_subject(fortinet-ca2)
Remote CA subject is /C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=fortinet-subca2001/emailAddress=support@fortinet.com.
issuer matching...try next if not match... local_issuer(fortinet-subca2001), remote_CA_subject(fortinet-subca2001)
CA issuer matched, local=remote=fortinet-subca2001
Find cert idx=0, peer_ca = 4
__use_cert,617: start idx = 0
use_certificate_issuer = /C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=fortinet-subca2001/emailAddress=support@fortinet.com
    subject = /C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=FortiManager/CN=FMG-VHTR26011676/emailAddress=support@fortinet.com
Use cert idx=0 by peer_ca = 4
__info_callback,993: role=svr,state=23, TLSv1.3 SSLv3/TLS write certificate
__info_callback,993: role=svr,state=40, TLSv1.3 TLSv1.3 write server certificate verify
__info_callback,993: role=svr,state=36, TLSv1.3 SSLv3/TLS write finished
__info_callback,993: role=svr,state=46, TLSv1.3 TLSv1.3 early data
__info_callback,993: role=svr,state=46, TLSv1.3 TLSv1.3 early data
TLSv1.3 write fatal alert: unknown
fw_proto_ssl.c,1026: TLSv1.3 error
fw_proto_ssl.c, __get_error,1615, err=167772359, error:0A0000C7:SSL routines:peer did not return a certificate.
fw_proto_ssl.c, __get_error,1629, ret=-4, error=1, errno=0, success.
proxy_session.c, proxy_session_cleanup, 118:cnt=0, session=0x55f5c0ebb00c.

```

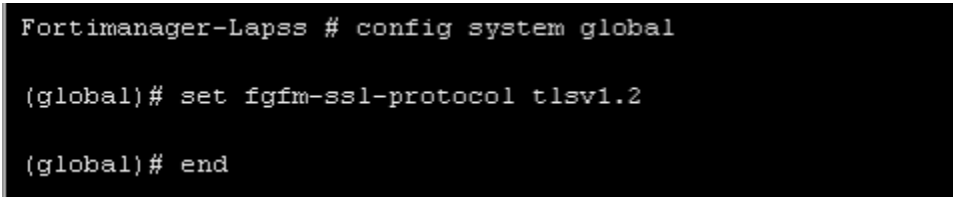
Figure 19 — FGFM debug — four CA issuers broadcast, and the handshake still failing.

The debug also shows the handshake reaching the certificate-request stage and then ending in a fatal alert, with FortiManager reporting that the peer did not return a certificate.

▸ 3.6 TLS version and chain pinning

A 512-bit RSA key cannot produce the RSA-PSS signatures that TLS 1.3 mandates, so FGFM has to be forced down to TLS 1.2 for a certificate that small to be usable at all.

```
config system global
    set fgfm-ssl-protocol tlsv1.2
end
```



```
Fortimanager-Lapss # config system global

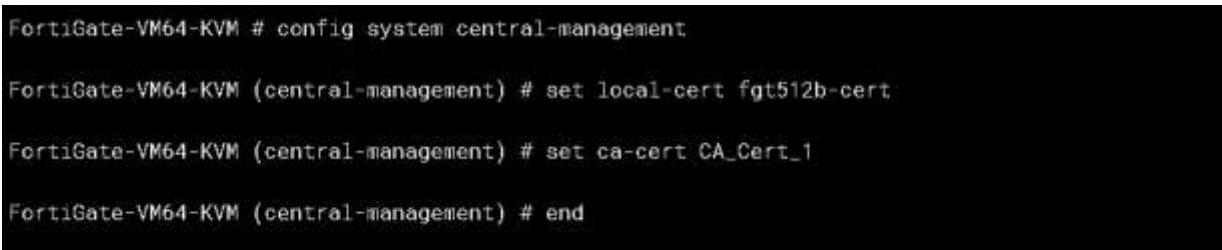
(global)# set fgfm-ssl-protocol tlsv1.2

(global)# end
```

Figure 20 — FGFM forced to TLS 1.2 to accommodate the 512-bit key.

On the FortiGate side both the device certificate and the CA were then pinned.

```
config system central-management
    set local-cert fgt512b-cert
    set ca-cert CA_Cert_1
end
```



```
FortiGate-VM64-KVM # config system central-management

FortiGate-VM64-KVM (central-management) # set local-cert fgt512b-cert

FortiGate-VM64-KVM (central-management) # set ca-cert CA_Cert_1

FortiGate-VM64-KVM (central-management) # end
```

Figure 21 — Device certificate and CA pinned on the FortiGate.

This reversed the direction of the failure. With ca-cert pinned, the FortiGate validates FortiManager's certificate against the lab CA only — and FortiManager presents its own Fortinet-signed certificate, which that CA did not issue. The FortiGate then rejects FortiManager for an untrusted chain. Removing the pin returns the failure to the FortiManager side.

Why this cannot be made to work

FortiManager requires a certificate whose CN contains the serial and whose issuer it trusts, negotiated over a modern TLS version. The unlicensed FortiGate will import only a 512-bit key, which cannot be used with TLS 1.3 and would be rejected as too weak by any current TLS implementation.

The only configuration that satisfies both at once is a 512-bit RSA chain over TLS 1.2 — a path through the requirements that no production network should ever run. The limitation is structural, not a misconfiguration.

04. Attempt 2 — Licensed FortiManager and FortiGate 7.6.7

If the ceiling on imported keys comes from the evaluation licence, licensing both appliances should remove it. A fresh pair was deployed at 7.6.7 and both were registered for trial entitlement.



Figure 22 — FortiGate and FortiManager 7.6.7 KVM images.

FGVMEV21T3HWS400	FortiGate VM Trial	No coverage	2026-10-05
FMG-VM3M656011750	FortiManager VM	No coverage	2026-10-05

Figure 23 — Both VMs registered in the support portal.

► 4.1 FortiManager-initiated registration

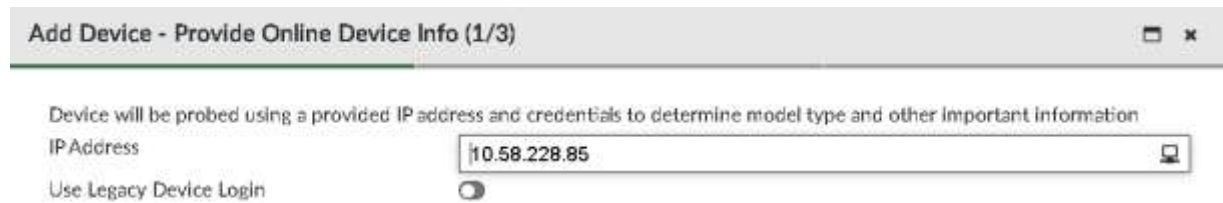


Figure 24 — Add Device > Discover Device probing 10.58.228.85.

The probe succeeded — on Pair A this step failed outright, so the licensed appliances clear the earlier TLS problem. FortiManager then waits for the FortiGate to approve the management relationship.



Figure 25 — Probe successful — waiting for authorization from the device.

On current FortiOS that approval is a Security Fabric authorization rather than a simple credential exchange, so the request surfaces in the FortiGate GUI and has to be allowed there.

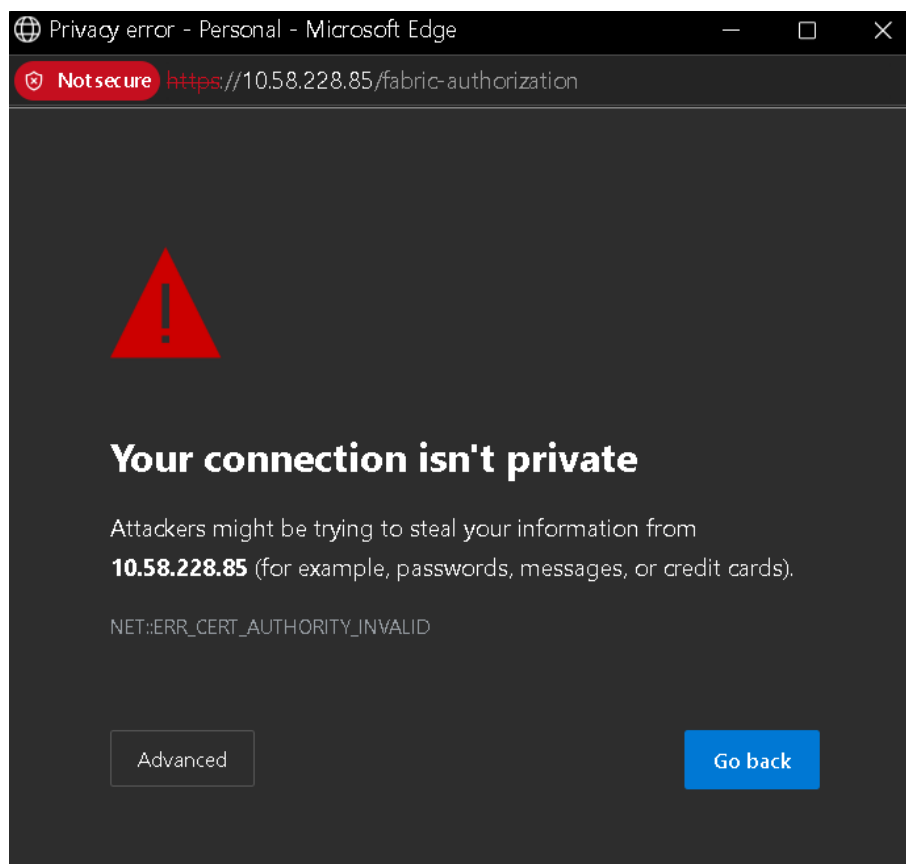


Figure 26 — Browser certificate warning reaching the fabric-authorization page over HTTPS.

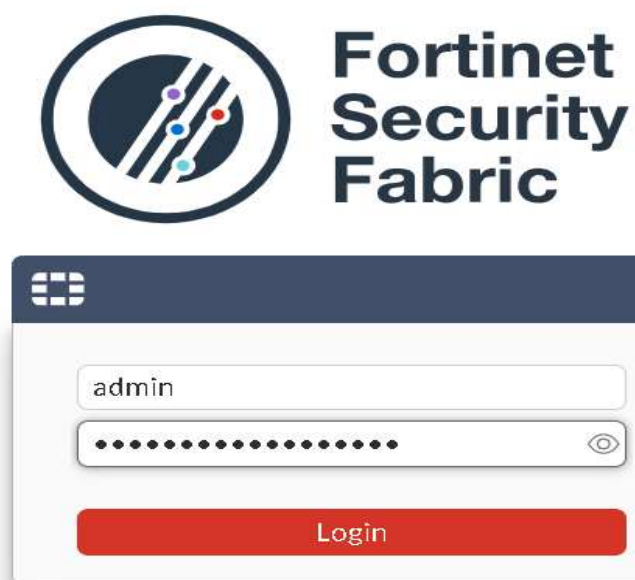


Figure 27 — Fabric authorization login.

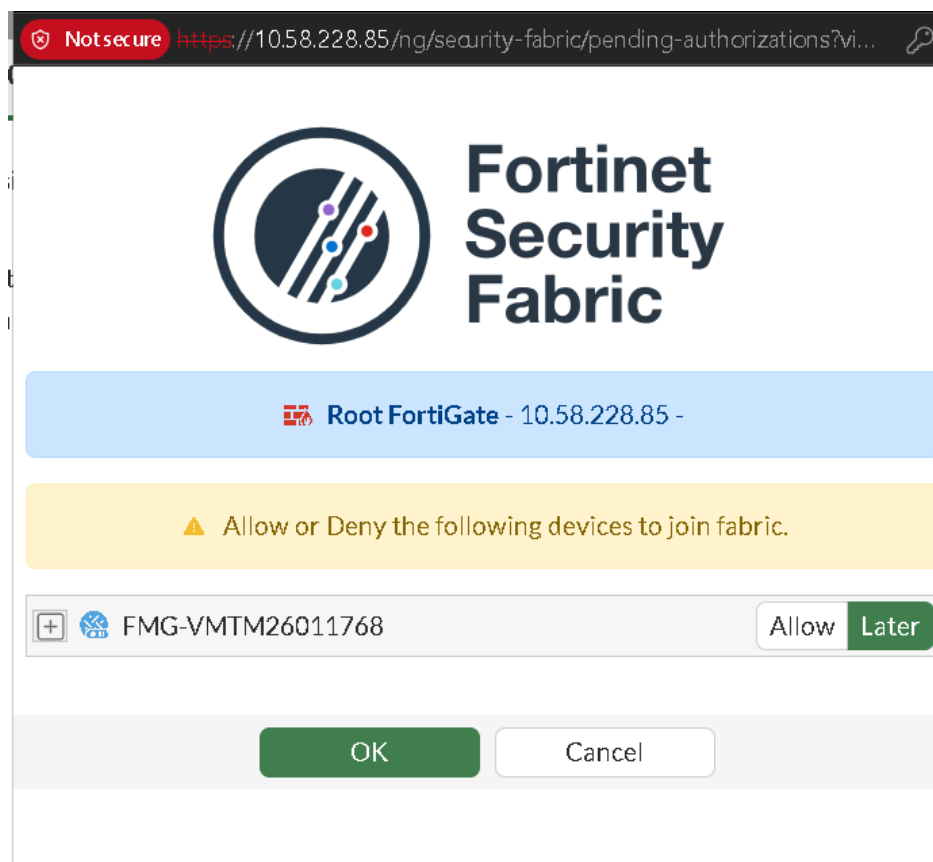


Figure 28 — Pending authorization — FortiManager FMG-VMTM26011768 requesting to join the fabric.

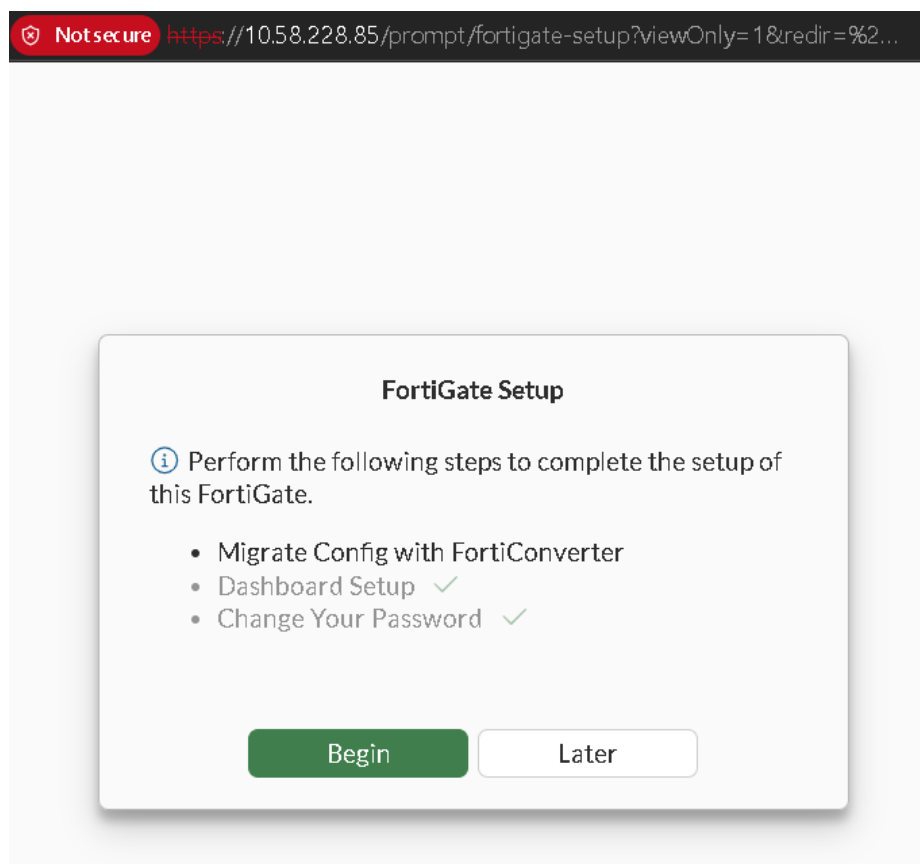


Figure 29 — FortiGate setup prompt presented alongside the authorization request.

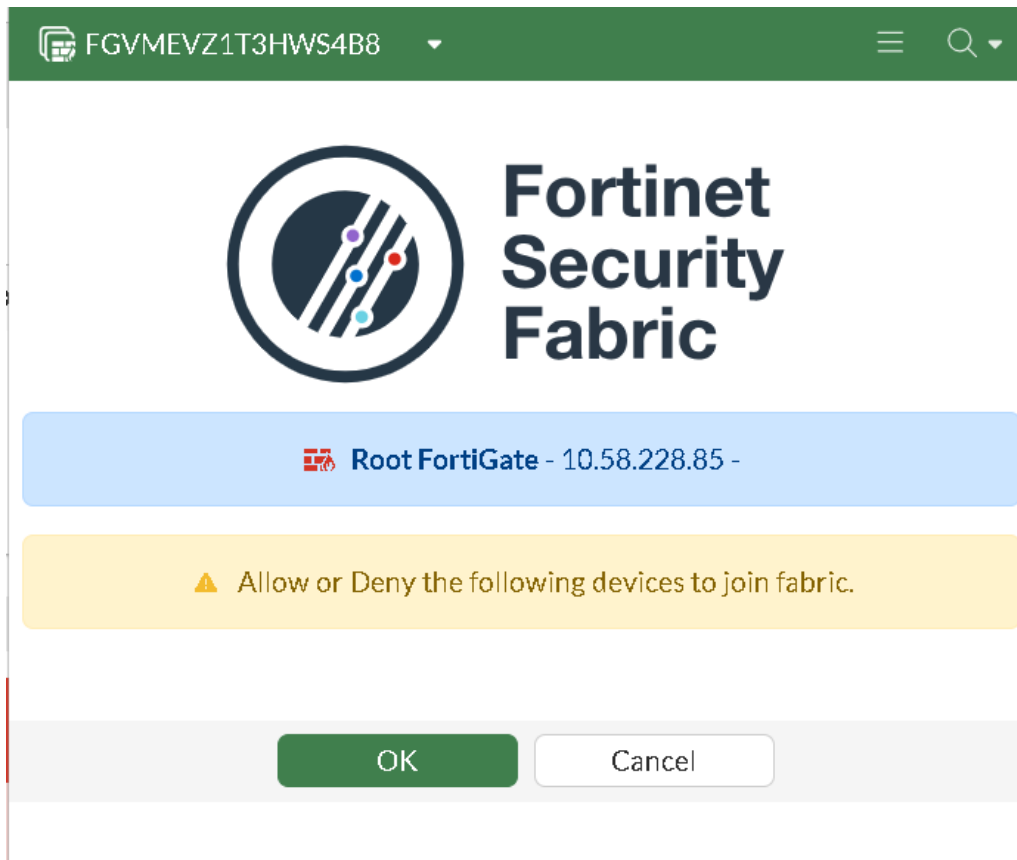


Figure 30 — Fabric authorization dialog as rendered inside the FortiGate GUI.



Figure 31 — The same fabric-authorization page reached over HTTP, to rule out the certificate warning.

► 4.2 Device-initiated registration

```
config system central-management
  set type fortimanager
  set fmg 10.58.228.84
end
```

The FortiGate warns that without a FortiManager serial number the connection is treated as unverified, offers to fetch it, and then fails to retrieve it.

```

FGVMEVZ1T3HWS4B8 # config system central-management
FGVMEVZ1T3HWS4B8 (central-management) # set type fortimanager
FGVMEVZ1T3HWS4B8 (central-management) # set fmg 10.58.228.84
FGVMEVZ1T3HWS4B8 (central-management) # end
The Serial Number for FortiManager is not entered.
In order to verify identity of FortiManager serial number is needed.
If serial number is not set, connection will be set as unverified.
FortiGate can establish a connection to obtain the serial number now.Do you want to try to connect now? (y/n)y
Failed to get FortiManager SN from 10.58.228.84.
Do you want to try to connect now? (y/n)y
Failed to get FortiManager SN from 10.58.228.84.
Do you want to try to connect now? (y/n)

```

Figure 32 — FortiGate unable to retrieve the FortiManager serial number.

Reachability is not the cause — ICMP succeeds and a TCP connection to port 541 is established.

```

FGVMEVZ1T3HWS4B8 # execute ping 10.58.228.84
PING 10.58.228.84 (10.58.228.84): 56 data bytes
64 bytes from 10.58.228.84: icmp_seq=0 ttl=64 time=2.8 ms
64 bytes from 10.58.228.84: icmp_seq=1 ttl=64 time=2.0 ms
64 bytes from 10.58.228.84: icmp_seq=2 ttl=64 time=1.7 ms
64 bytes from 10.58.228.84: icmp_seq=3 ttl=64 time=2.0 ms
64 bytes from 10.58.228.84: icmp_seq=4 ttl=64 time=1.8 ms

--- 10.58.228.84 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.7/2.0/2.8 ms

FGVMEVZ1T3HWS4B8 # execute telnet 10.58.228.84 541
Trying 10.58.228.84...
Connected to 10.58.228.84.

```

Figure 33 — Ping and telnet to port 541 both succeeding.

► 4.3 What the FGFM debug shows

The debug is unambiguous: SSL negotiation finishes successfully, and the FortiGate sends its registration message carrying the serial, platform, firmware and management address. The transport layer is healthy; the relationship is refused above it.

```
FGFMs: SSLv3/TLS write finished
FGFMs: SSL negotiation finished successfully
FGFMs: client:send:
get auth
serialno=FGVMEVZ1T3HWS4B8
mgmtid=00000000-0000-0000-0000-000000000000
platform=FortiGate-VM64-KVM
fos_ver=700
minor=6
patch=7
build=3704
branch=3704
maxvdom=2
fg_ip=10.58.228.85
hostname=FGVMEVZ1T3HWS4B8
harddisk=no
biover=04000002
mgmt_mode=normal
enc_flags=0
mgmtip=10.58.228.85
mgmtport=443

FGFMs: SSL negotiation finished successfully
FGFMs: SSL negotiation finished successfully
FGFMs: SSLv3/TLS read server session ticket
FGFMs: SSL negotiation finished successfully
FGFMs: SSL negotiation finished successfully
FGFMs: SSLv3/TLS read server session ticket
FGFMs: Cleanup session 0x55bc54ec2c90, 10.58.228.84.
FGFMs: Destroy session 0x55bc54ec2c90, 10.58.228.84.
```

Figure 34 — FGFM debug — SSL negotiation successful and the get auth message sent.

```

FGFMs: SSLv3/TLS write finished
FGFMs: SSL negotiation finished successfully
FGFMs: client:send:
get auth
serialno=FGVMEVZ1T3HWS488
mgmtid=00000000-0000-0000-0000-000000000000
platform=FortiGate-VM64-KVM
fos_ver=700
minor=6
patch=7
build=3704
branch=3704
maxvdom=2
fg_ip=10.58.228.85
hostname=FGVMEVZ1T3HWS488
harddisk=no
biover=04000002
mgmt_mode=normal
enc_flags=0
mgmtip=10.58.228.85
mgmtport=443

FGFMs: SSL negotiation finished successfully
FGFMs: SSL negotiation finished successfully
FGFMs: SSLv3/TLS read server session ticket
FGFMs: SSL negotiation finished successfully
FGFMs: SSL negotiation finished successfully
FGFMs: SSLv3/TLS read server session ticket
FGFMs: Cleanup session 0x55bc54ec2c90, 10.58.228.84.
FGFMs: Destroy session 0x55bc54ec2c90, 10.58.228.84.

```

Figure 35 — The same exchange captured on the FortiManager side.

FortiManager can be told to accept registration requests from devices whose serial it does not already hold, which rules out the unknown-serial case as the cause.

```

config system global
    set fgfm-deny-unknown disable
end

```

```

fortimanager-test # config system global

(global)# set fgfm-deny-unknown
disable    Allow devices with unknown SN to actively register as an unauthorized device.
enable     Deny devices with unknown SN to actively register as an unauthorized device.

(global)# set fgfm-deny-unknown disable

(global)# next
Unknown action 0

(global)# end

```

Figure 36 — `fgfm-deny-unknown` disabled on FortiManager.

Licensing and certificate naming are separate limitations

The factory certificate on the licensed 7.6.7 FortiGate still carries CN = FortiGate rather than the serial number, and FortiManager rejects it exactly as it rejected the unlicensed device.

What the licence does lift is the key-size ceiling from section 3.4. These are two problems that looked like one on the evaluation VM: the crypto restriction comes from the licence, the certificate naming does not.

05. Attempt 3 — Version Pairing

► 5.1 Why a version change resolves it

FortiManager has not always enforced the serial-in-CN check. It was hardened in 7.0.12, 7.2.5 and 7.4.3, and those releases included an explicit opt-out for environments that could not meet it. That opt-out was then removed.

you can see:

Note:

Starting in v7.6.2/v7.4.8/v7.2.10, the `fgfm-peercert-withoutsn` command is no longer supported, and FortiManager will always check the FortiGate's certificate for the FortiGate Serial Number.

You can see that the command that allow register fortigate in fortimanager without certificates is only valid in lower versions:

- From v7.2.10/v7.4.6/v7.6.2 the above command is not available anymore.

Included versions

In summary, you can use this command only until versions 7.2.9, 7.4.5 and 7.6.1

```
config sys global
    set fgfm-peercert-withoutsn enable
```

And this command is the key for register Eval VM (kvm) fortigate in Fortimanager Eval VM (kvm).

Figure 37 — Fortinet documentation — `fgfm-peercert-withoutsn` removed from 7.2.10, 7.4.6 and 7.6.2.

FortiManager release	Serial-in-CN check	fgfm-peercert-withoutsu	Evaluation VM can register
Before 7.0.12 / 7.2.5 / 7.4.3	Not enforced	Not required	Yes
7.0.12 / 7.2.5 / 7.4.3 onward	Enforced	Available as opt-out	Yes, with the opt-out set
7.2.10 / 7.4.6 / 7.6.2 onward	Enforced	Removed	No

FortiManager 7.6.2, the appliance used in attempts 1 and 2, is past the removal, which is why no command existed to relax the check. FortiManager 7.4.3 still has it. Pairing that release with FortiOS 7.2.1, which predates the corresponding device-side change, produces a combination in which registration proceeds.

▸ 5.2 Environment preparation

```

config router static
  edit 1
    set dst 0.0.0.0 0.0.0.0
    set gateway 10.58.228.107
    set device port1
  next
end

config system dns
  set primary 8.8.8.8
  set secondary 1.1.1.1
end

```

```

FortiGate-VM64-KVM # config router static
FortiGate-VM64-KVM (static) # edit 1
new entry '1' added

FortiGate-VM64-KVM (1) # set dst 0.0.0.0 0.0.0.0
FortiGate-VM64-KVM (1) # set gateway 10.58.228.107
FortiGate-VM64-KVM (1) # set device port1

FortiGate-VM64-KVM (1) # next
The destination is set to 0.0.0.0/0 which means all IP addresses.

FortiGate-VM64-KVM (static) # end

FortiGate-VM64-KVM # config system dns
FortiGate-VM64-KVM (dns) # set primary 8.8.8.8
FortiGate-VM64-KVM (dns) # set secondary 1.1.1.1

```

Figure 38 — Default route and DNS configured on the FortiGate.

System Information

Host Name	FMG-VM64	
Serial Num...	FMG-VMTM26011676	
Platform Ty...	FMG-VM64	
HA Status	Standalone	
System Time	Wed Oct 07 07:23:13 2026 PDT	
Firmware V...	v7.4.3-build2487 240514 (GA)	
System Co...	Last Backup: N/A	
Current Ad...	admin / 2 in total	
Up Time	16 minutes 4 seconds	

Figure 39 — FortiManager 7.4.3 build2487.

System Information	
Hostname	fortigate7-2
Serial Number	FGVMEV6EP-2D3VB4
Firmware	v7.2.1 build1254 (Feature)
Mode	NAT
System Time	2026/10/07 09:57:25
Uptime	00:00:00:56
WAN IP	Unknown

Figure 40 — FortiGate 7.2.1 build1254, hostname fortigate7-2, serial FGVMEV6EP-2D3VB4.

► 5.3 Relaxing the certificate check

On FortiManager, the peer-certificate serial requirement is disabled. This is the single change that makes the difference, and it is available here only because of the firmware version.

```
config system global
    set fgfm-peercert-withoutsn enable
end
```

```
Connected
FMG-VM64 # config system global

(global)# set fgfm-peercert-withoutsn enable

(global)# end

FMG-VM64 #
```

Figure 41 — fgfm-peercert-withoutsn enabled on FortiManager 7.4.3.

A lab accommodation, not a deployment recommendation

This command exists in these releases precisely because it weakens an authentication check, and it was removed for that reason.

In production the condition does not arise: an entitled FortiGate's factory certificate carries its serial in the subject CN and is signed by a Fortinet sub-CA, so both tests in section 3.1 pass without intervention. Where a device genuinely cannot present such a certificate, the supported route is an organisation PKI with full-size keys, trusted by FortiManager through fgfm-ca-cert.

▸ 5.4 Central management on the FortiGate

```
config system central-management
    set mode normal
    set type fortimanager
    set fmg "10.58.228.15"
    set allow-push-configuration enable
    set allow-push-firmware enable
    set allow-remote-firmware-upgrade enable
    set allow-monitor enable
    set schedule-config-restore enable
    set schedule-script-restore enable
end
```

```

fortigate7-2 # config system central-management
fortigate7-2 (central-management) # set mode normal
fortigate7-2 (central-management) # set type fortimanager
fortigate7-2 (central-management) # set schedule-config-restore enable
fortigate7-2 (central-management) # set schedule-script-restore enable
fortigate7-2 (central-management) # set allow-push-configuration enable
fortigate7-2 (central-management) # set allow-push-firmware enable
fortigate7-2 (central-management) # set allow-remote-firmware-upgrade enable
fortigate7-2 (central-management) # set allow-monitor enable
fortigate7-2 (central-management) # set serial-number "FGWMEV6EP-203V64"
command parse error before 'serial-number'
Command fail. Return code -61

fortigate7-2 (central-management) # set fq "10.88.228.15"
fortigate7-2 (central-management) # set fq-source-ip 8.8.8.8
fortigate7-2 (central-management) # set fq-source-ip0 ::
fortigate7-2 (central-management) # set local-cert ''
fortigate7-2 (central-management) # unset ca-cert
fortigate7-2 (central-management) # set vdan "root"
fortigate7-2 (central-management) # set fq-update-port 8890
fortigate7-2 (central-management) # set include-default-servers enable
fortigate7-2 (central-management) # set enc-algorithm default
fortigate7-2 (central-management) # set interface-select-method auto
fortigate7-2 (central-management) # end

```

Figure 42 — Central management configured on fortigate7-2.

Observation	set serial-number returned command parse error before 'serial-number', return code -61.
Why	The attribute is not valid under config system central-management on FortiOS 7.2.1, and it is not needed — the serial is carried in the FGFM registration message itself, as the debug in Figure 34 shows.
Effect	The remaining commands in the block applied normally. A parse error part-way through a block does not abort the rest of it.

► 5.5 The device appears as unauthorized

This is the state the earlier attempts never reached. The FortiGate has established FGFM, FortiManager has read its name, platform, serial, address and firmware from the registration message, and it is being held pending approval.

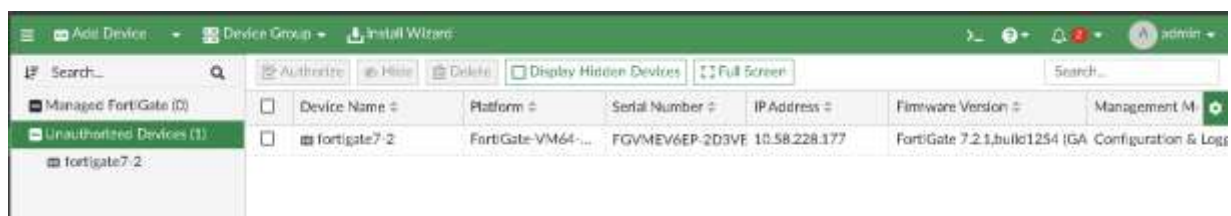


Figure 43 — fortigate7-2 listed under Unauthorized Devices with its details read correctly.

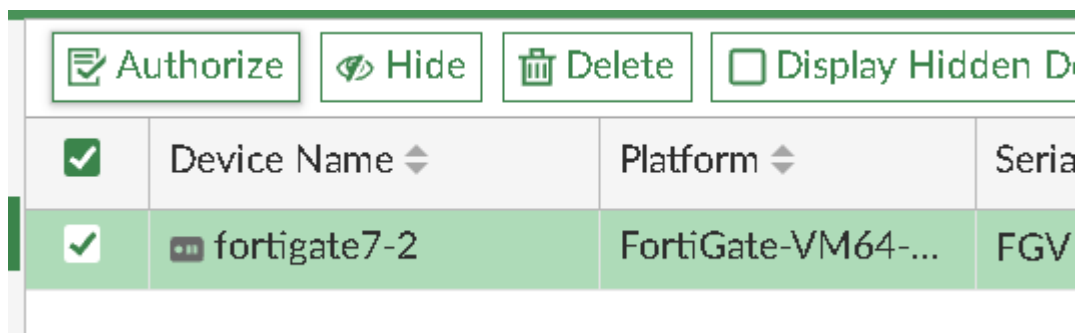


Figure 44 — Selecting the device and authorizing it.

5.6 ADOM firmware matching

Authorization asks which ADOM the device joins. The default offered is root at FortiGate 7.4, which does not match a 7.2.1 device — a device can only be managed in an ADOM of its own firmware version, and a mismatch here is one of the most common causes of a registration that appears to fail for no reason.

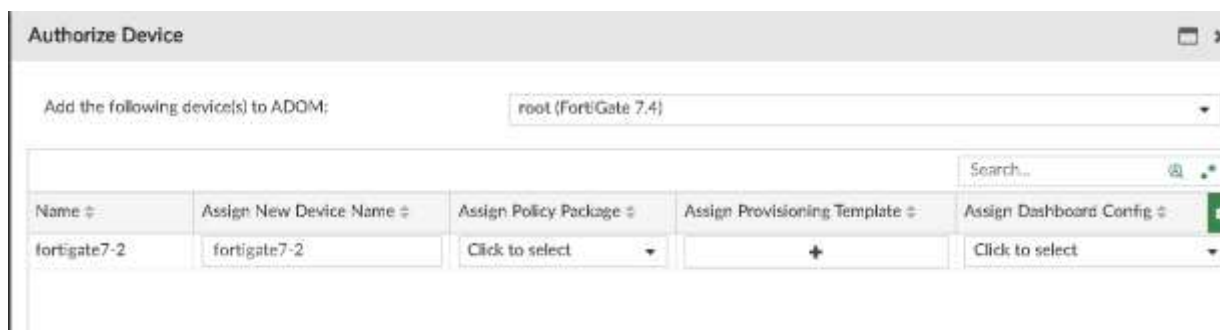


Figure 45 — Authorize Device defaulting to root (FortiGate 7.4).

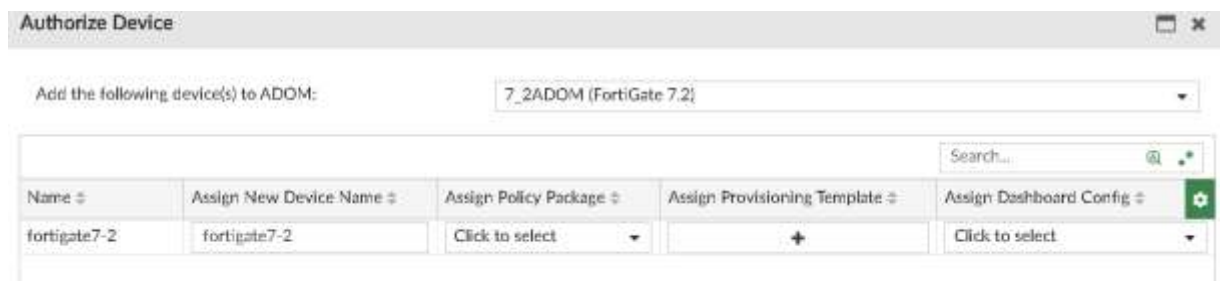


Figure 46 — Device assigned to 7_2ADOM (FortiGate 7.2).

▸ 5.7 Completing through the Discover wizard

The device was also taken through the Discover Device wizard, which reads its details directly rather than waiting for a registration message. The information shown here was obtained from the device itself, not supplied by hand.



Add Device - Discover Device (2/3)

The following information has been discovered from the device:

IP Address	10.58.228.177
Host Name	fortigate7-2
SN	FGVMEV6EP-2D3VB4
Model	FortiGate-VM64-KVM
Firmware Version	7.2.1, build 1254 (GA)
HA Status	Standalone
Administrator	admin

Please input the following information to complete addition of the device:

Name	fortigate7-2
Description	Description
Provisioning Templates	
Add To Folder	☒
Add To Device Group	☒
Copy Device Dashboard	Click to select

Figure 47 — Discover Device — model, serial, firmware and HA status read from the FortiGate.

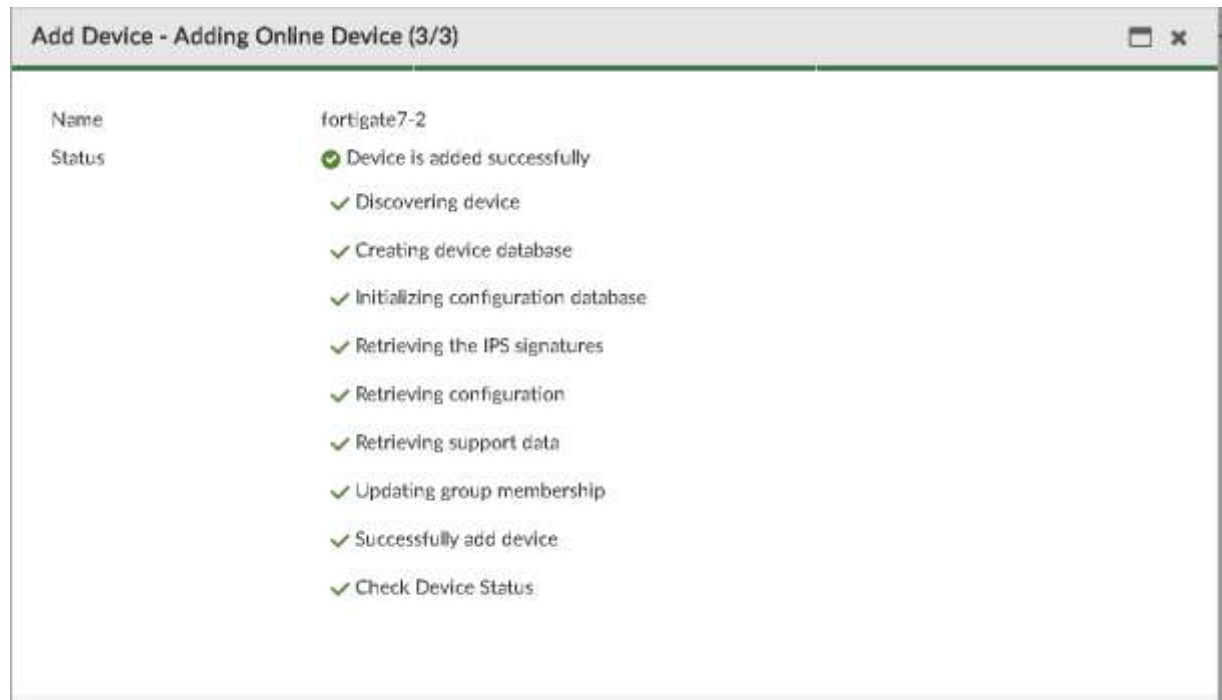


Figure 48 — Device added — database created, configuration and support data retrieved.

► 5.8 Managed state

Device Manager shows one managed FortiGate with connectivity green and configuration status Synchronized. The policy package ring stays amber at Never Installed, which is correct — registration establishes management, it does not install policy.

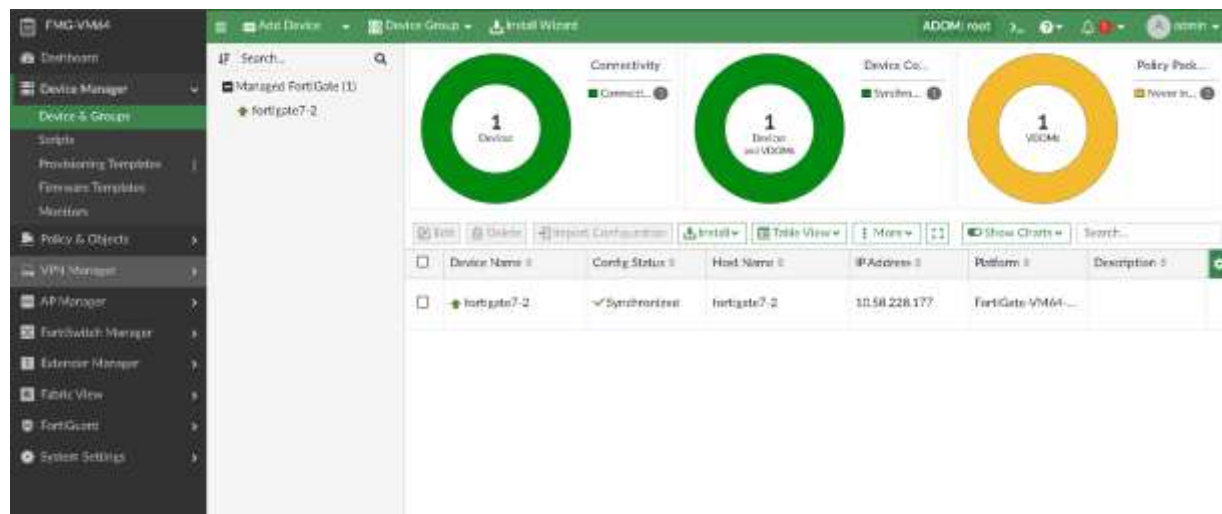


Figure 49 — fortigate7-2 managed and synchronized in Device Manager.

▸ 5.9 What managed mode changes on the FortiGate

Once a FortiGate is managed, logging in to it directly is no longer neutral. It warns that changes should be made from FortiManager and offers a read-only session, because a local change made outside FortiManager puts the device out of sync with the stored configuration.

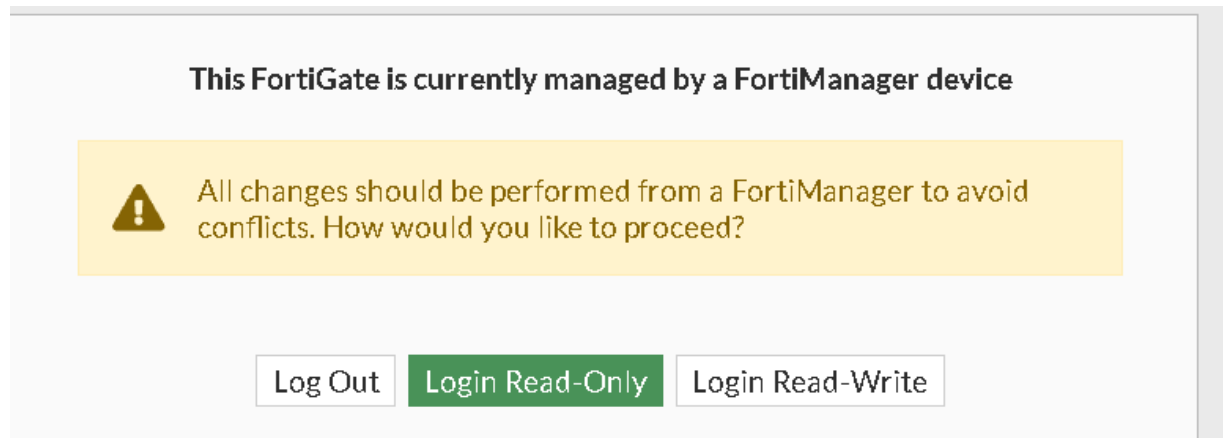


Figure 51 — FortiGate login warning that the device is under FortiManager management.

The supported way to reach the device is from FortiManager itself, through the Remote Access option on the device record, which tunnels the session over the management channel.



Figure 52 — Remote Access on the managed device record.